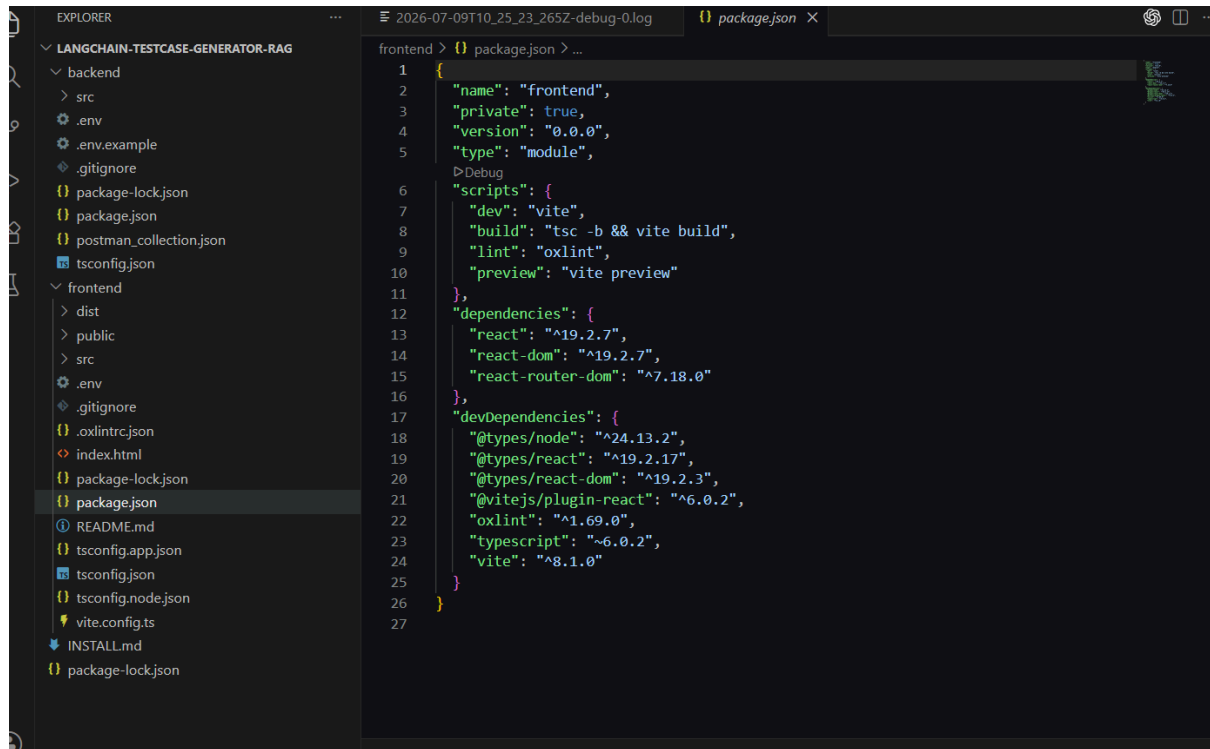


LANGCHAIN RAG Implementation

Import the below project in screenshot in vs code from a folder

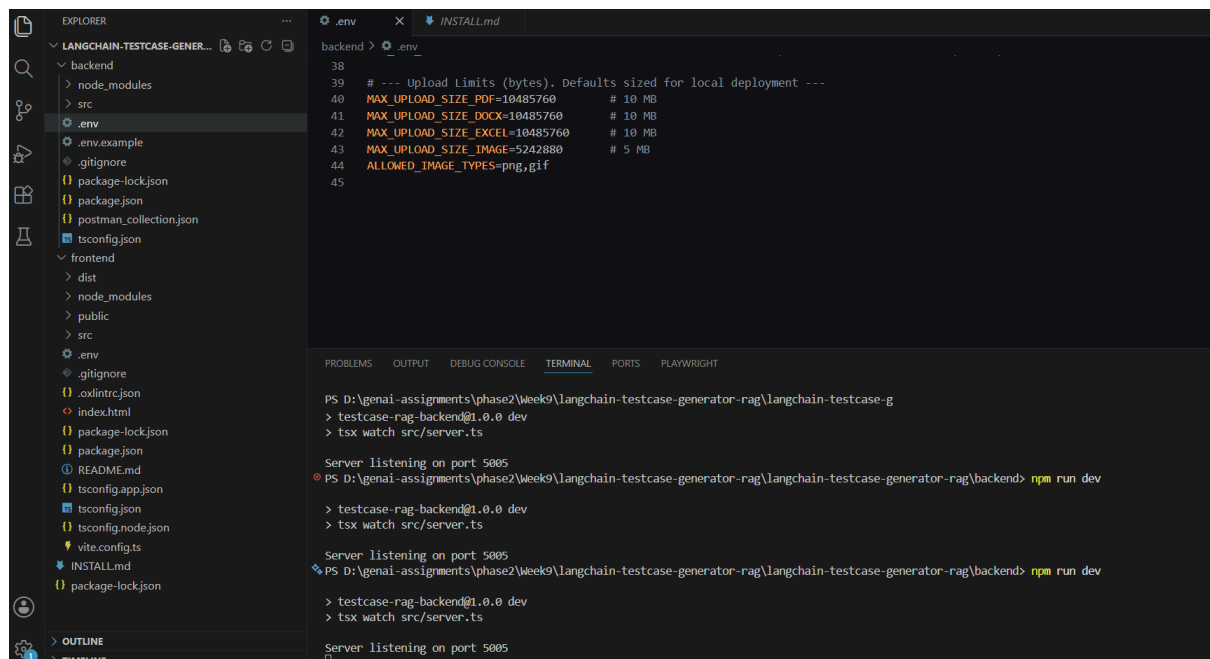


Run

cd D:\genai-assignments\phase2\Week9\langchain-testcase-generator-rag\backend

npm install

npm run dev



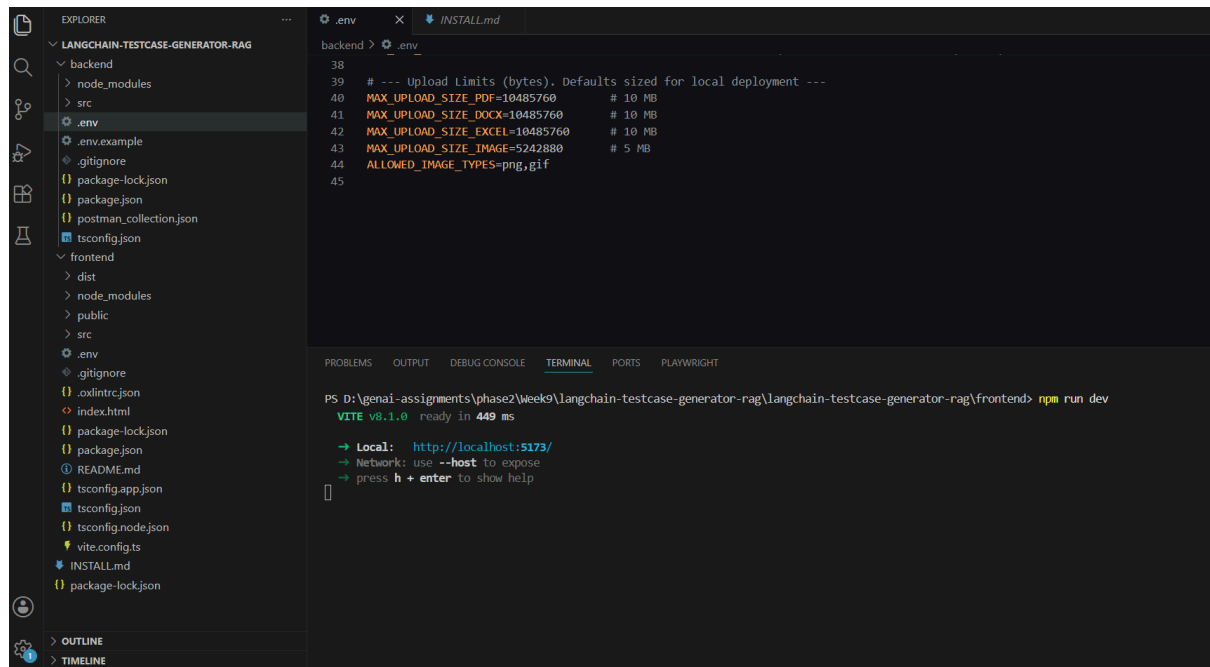
Now do the same for front end

Run

cd D:\genai-assignments\phase2\Week9\langchain-testcase-generator-rag\frontend

npm install

npm run dev



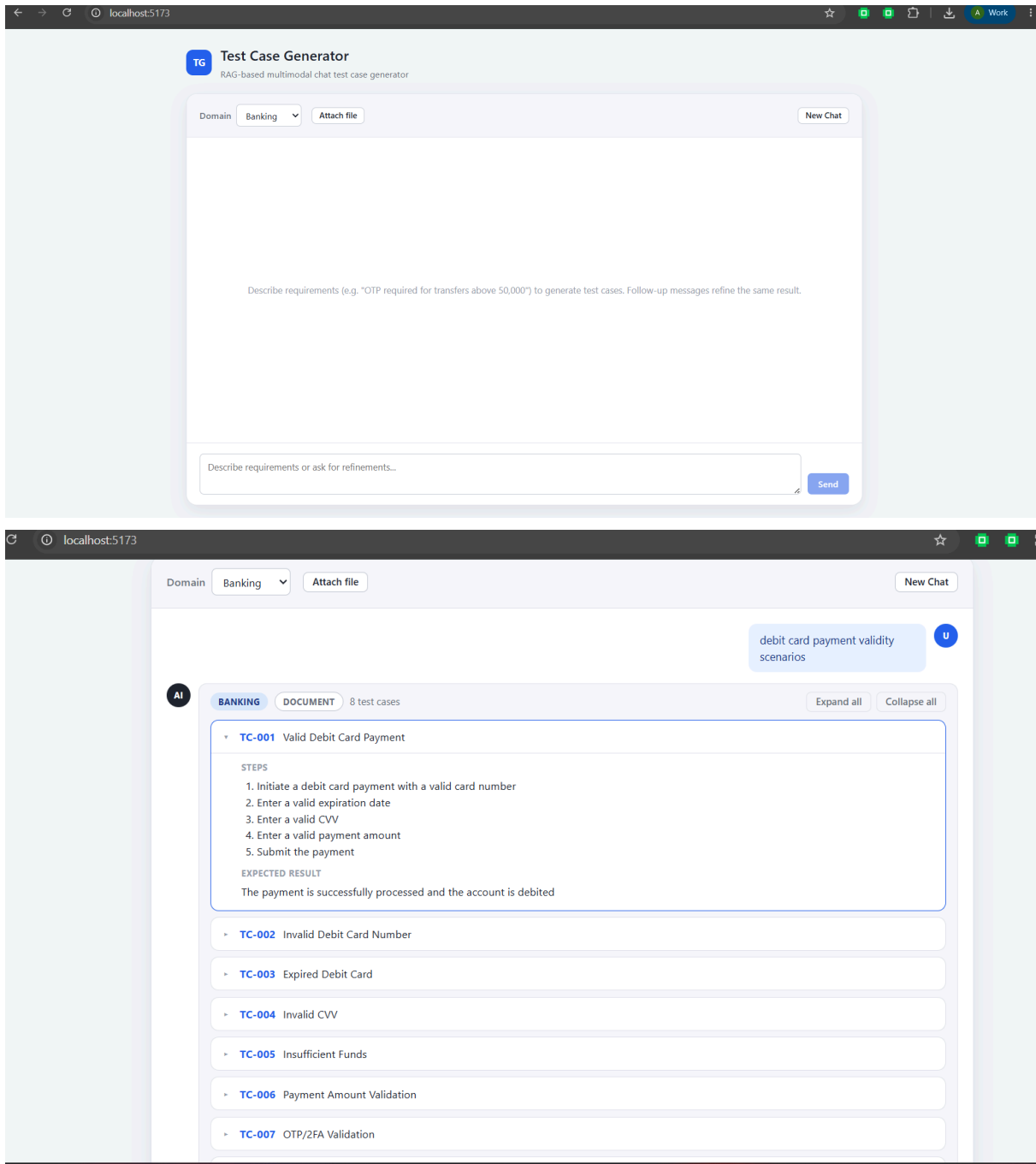
The screenshot shows the Visual Studio Code interface. On the left, the Explorer sidebar displays the project structure for 'LANGCHAIN-TESTCASE-GENERATOR-RAG', with the 'frontend' directory selected. The main editor area shows the '.env' file with the following content:

```
38
39 # --- Upload Limits (bytes). Defaults sized for local deployment ---
40 MAX_UPLOAD_SIZE_PDF=10485760 # 10 MB
41 MAX_UPLOAD_SIZE_DOCX=10485760 # 10 MB
42 MAX_UPLOAD_SIZE_EXCEL=10485760 # 10 MB
43 MAX_UPLOAD_SIZE_IMAGE=5242880 # 5 MB
44 ALLOWED_IMAGE_TYPES=png,gif
45
```

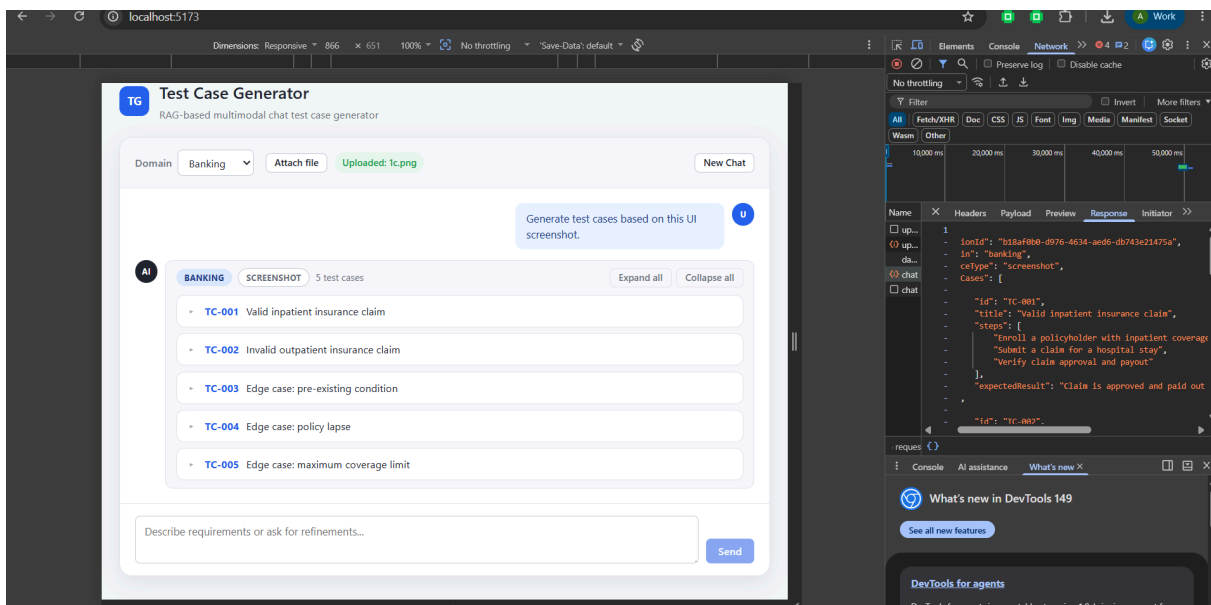
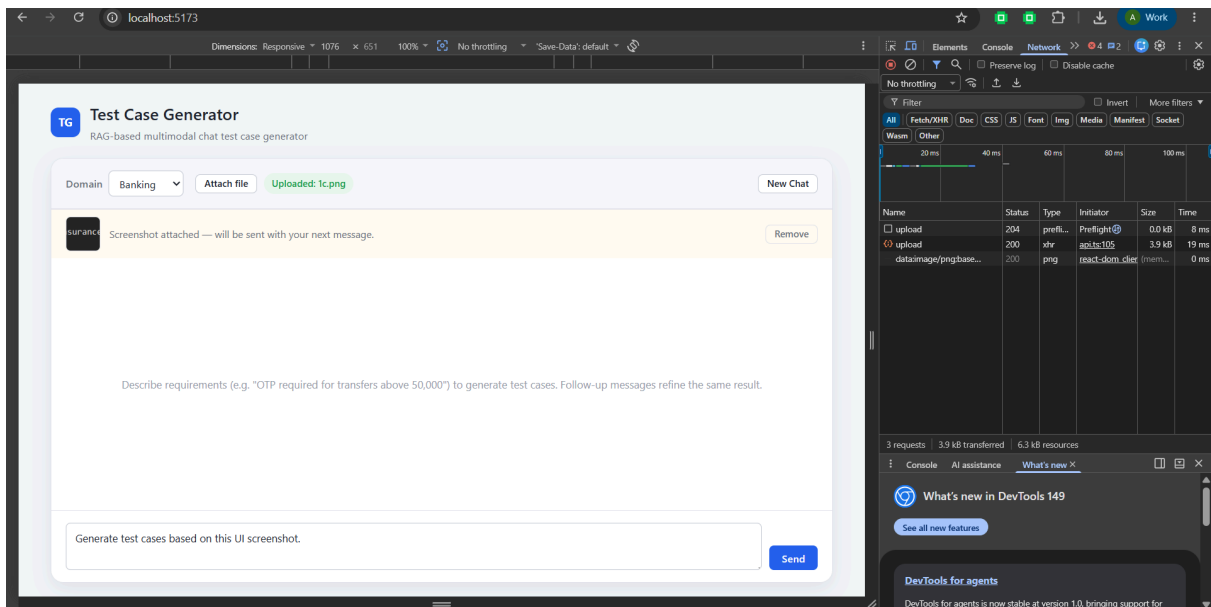
At the bottom, the Terminal panel shows the command prompt output for running 'npm run dev' in the frontend directory:

```
PS D:\genai-assignments\phase2\Week9\langchain-testcase-generator-rag\frontend> npm run dev
VITE v8.1.0 ready in 449 ms

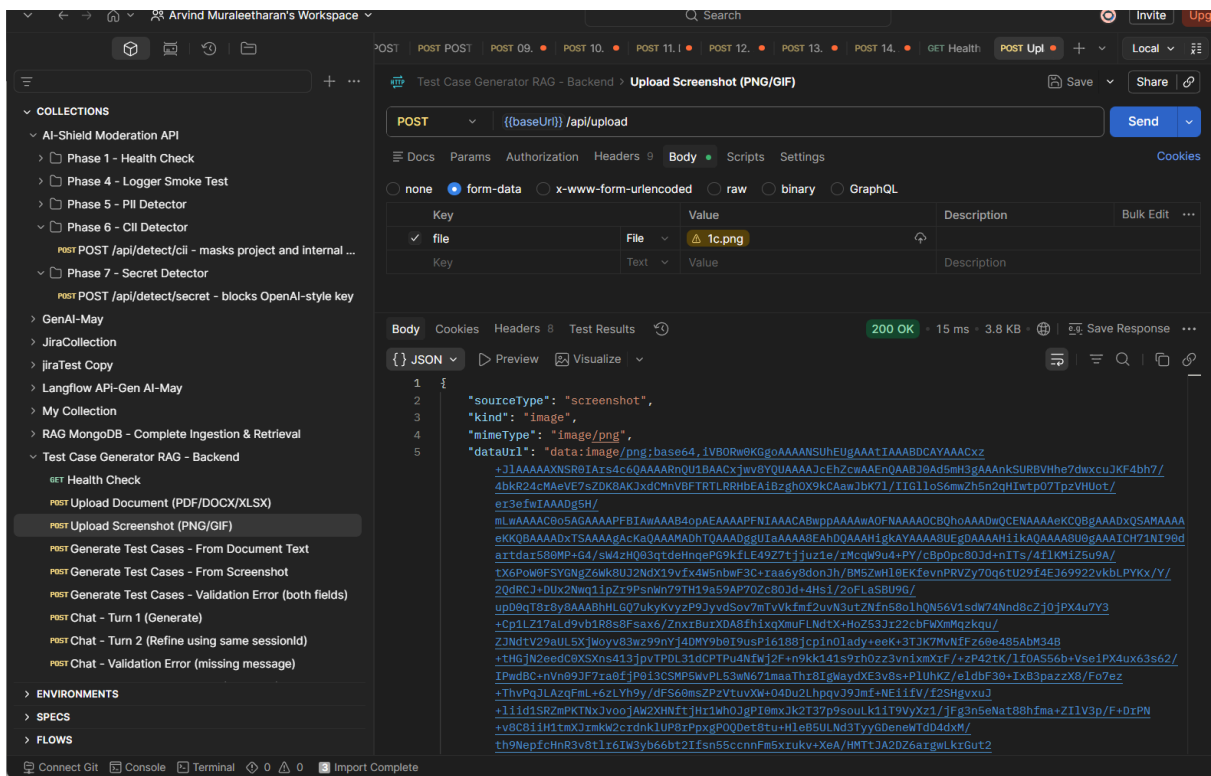
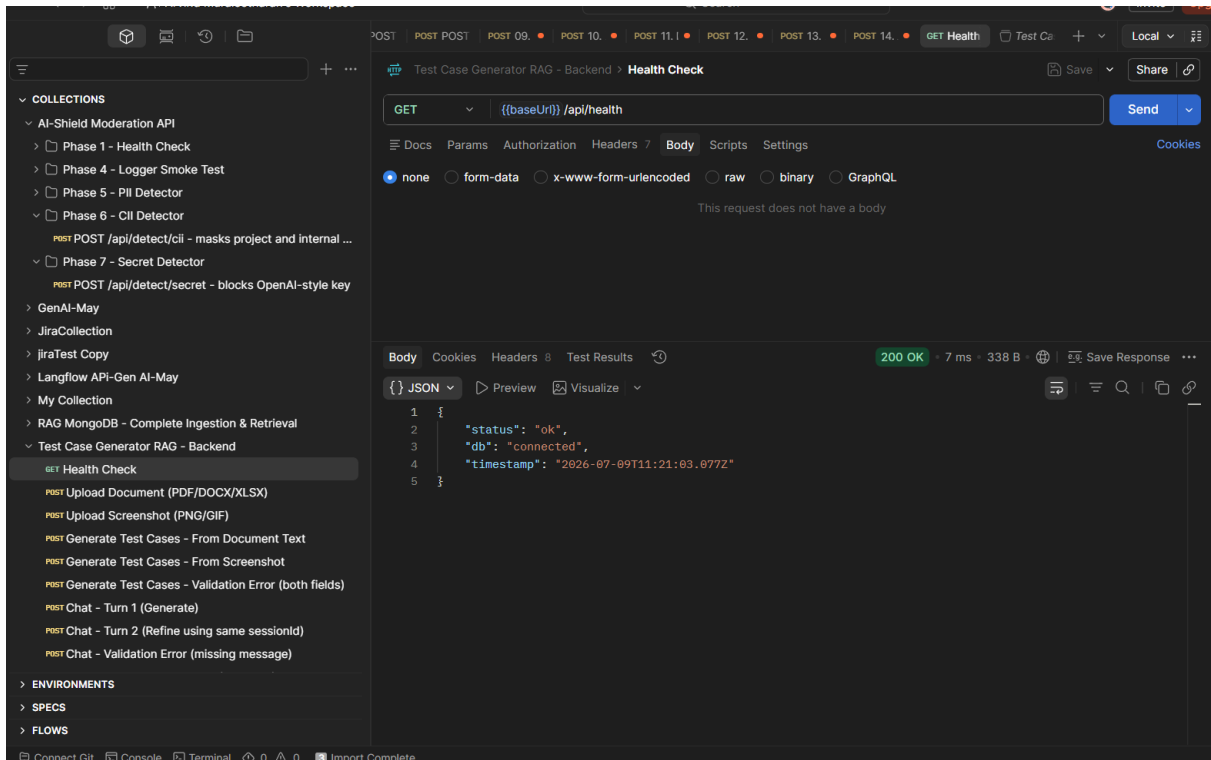
  Local: http://localhost:5173/
  Network: use --host to expose
  press h + enter to show help
```

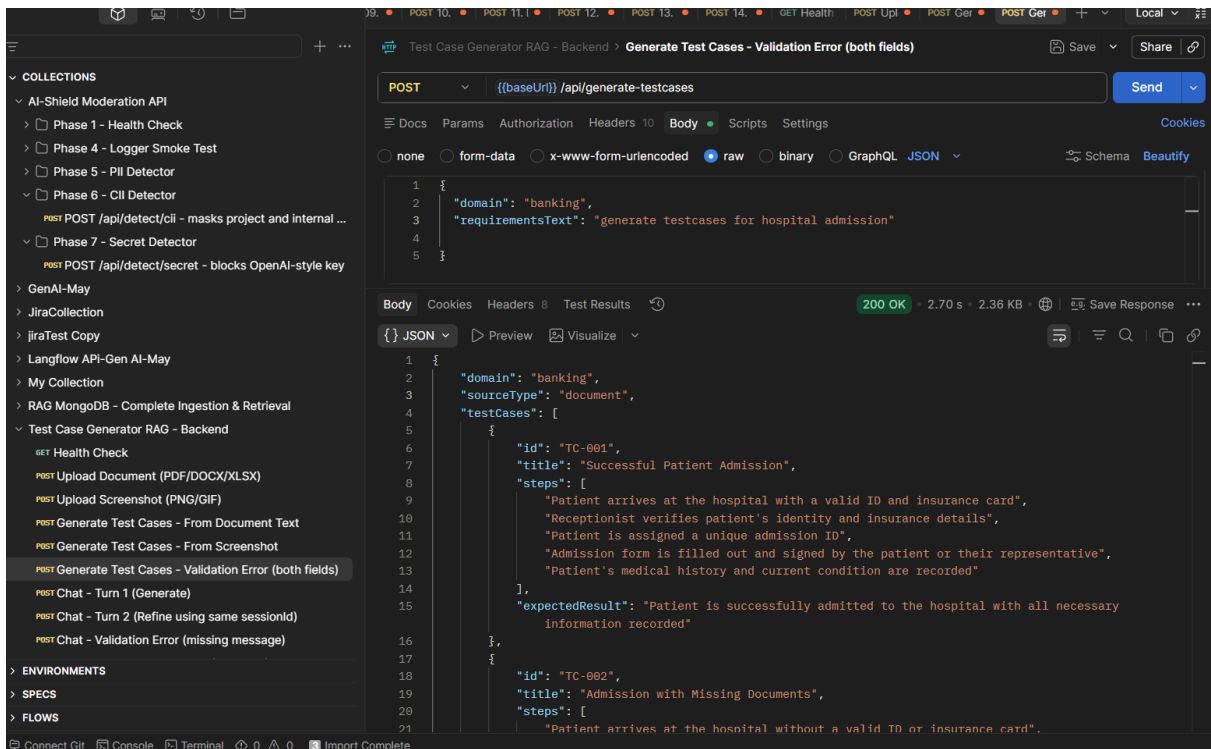
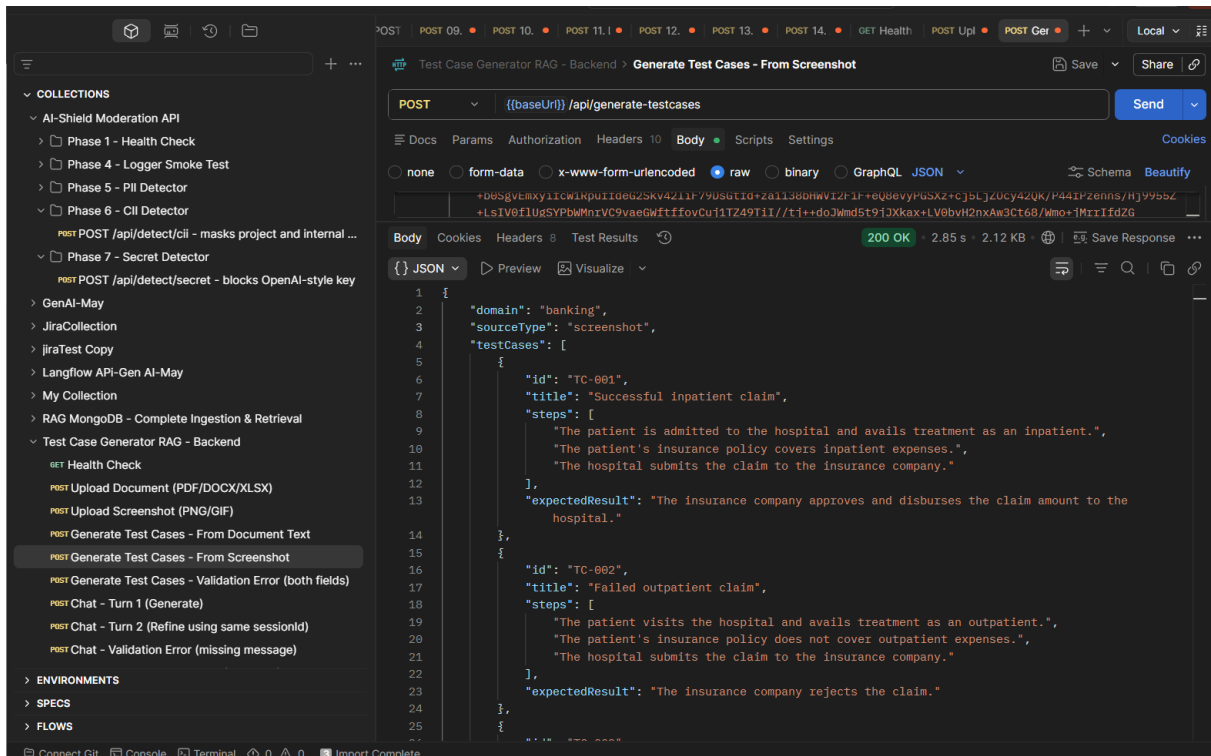


Test cases generation using screenshot



Postman collection verifications





Test Case Generator RAG - Backend > Chat - Turn 1 (Generate)

POST {{baseUrl}} /api/chat

Body

```
1 {
2   "domain": "insurance",
3   "message": "Claims above the coverage limit must be auto-flagged for manual underwriter review. Premium
4     must be recalculated whenever coverage is modified mid-term."
5 }
```

200 OK · 2.36 s · 2.01 KB

JSON

```
1 {
2   "sessionId": "bb200ca8-f15a-4eed-903c-eabe64728a52",
3   "domain": "insurance",
4   "sourceType": "document",
5   "testCases": [
6     {
7       "id": "TC-001",
8       "title": "Auto-flag claims above coverage limit",
9       "steps": [
10        "Create a new policy with a specified coverage limit",
11        "Submit a claim that exceeds the coverage limit",
12        "Verify the claim is auto-flagged for manual underwriter review"
13      ],
14      "expectedResult": "The claim is successfully flagged and pending manual underwriter review"
15    },
16    {
17      "id": "TC-002",
18      "title": "Premium recalculation after mid-term coverage modification",
19      "steps": [
20        "Create a new policy with a specified coverage limit and premium",
21        "Modify the coverage limit mid-term",

```

Test Case Generator RAG - Backend > Chat - Turn 2 (Refine using same sessionId)

POST {{baseUrl}} /api/chat

Body

```
1 {
2   "sessionId": "bb200ca8-f15a-4eed-903c-eabe64728a52",
3   "domain": "insurance",
4   "message": "Add one additional edge-case test for a claim filed on the policy expiration date."
5 }
```

200 OK · 1.32 s · 806 B

JSON

```
1 {
2   "sessionId": "bb200ca8-f15a-4eed-903c-eabe64728a52",
3   "domain": "insurance",
4   "sourceType": "document",
5   "testCases": [
6     {
7       "id": "TC-001",
8       "title": "Edge case - Claim filed on policy expiration date",
9       "steps": [
10        "Create a new policy with a specified coverage limit and expiration date",
11        "Submit a claim on the policy expiration date",
12        "Verify the system handles the claim and determines coverage based on the policy terms"
13      ],
14      "expectedResult": "The system handles the claim and determines coverage based on the policy
15        terms, considering the expiration date"
16    },
17  ]
18 }
```